

## Statistics and Probability (STA301)

Question: 1 (Marks: 1)

Attempted Questions: 4 Total Questions: 23

A five-number summary consists of :

## Choices:

☒  $X_0$ ,  $Q_1$ , Median,  $Q_3$ , and  $X_m$ *correct*☐  $X_m$ ,  $Q_1$ , Mean,  $Q_3$ , and  $X_0$ ☐  $X_m$ ,  $Q_1$ , Mode,  $Q_3$ , and  $X_0$ ☐  $X_0$ ,  $Q_1$ , Median,  $Q_2$ , and  $X_m$

## Statistics and Probability (STA301)

Question: 3 (Marks: 1)

Attempted Questions: 4

Total Questions: 23

In a multiplication theorem  $P(A \cap B)$  equals:

## Choices:

☒  $P(A) P(B)$

*correct*

☐  $P(A) + P(B)$

☐  $P(A) * P(B|A)$

☐  $P(B|A) * P(B)$

## Statistics and Probability (STA301)

Question: 4 (Marks: 1)

Attempted Questions: 4

Total Questions: 23

The probability of drawing a 'white' ball from a bag containing 4 red, 8 black and 3 white balls is:

## Choices:

☐ 0☐  $\frac{3}{15}$ *correct*☐  $\frac{1}{12}$ ☒  $\frac{1}{2}$

## Statistics and Probability (STA301)

Question: 5 (Marks: 1)

Attempted Questions: 4

Total Questions: 23

If  $\hat{y}$  is the predicted value for a given  $x$ -value and  $b$  is the  $y$ -intercept then the equation of a regression line for an independent variable  $x$  and a dependent variable  $y$  is:

## Choices:

☒  $\hat{y} = mx + b$ , where  $m = \text{slope}$

*correct*

☐  $x = \hat{y} + mb$ , where  $m = \text{slope}$

☐  $\hat{y} = x/m + b$ , where  $m = \text{slope}$

☐  $\hat{y} = x + mb$ , where  $m = \text{slope}$

## Statistics and Probability (STA301)

Question: 6 (Marks: 1)

Attempted Questions: 5

Total Questions: 23

If  $A$  and  $B$  are mutually exclusive events then  $P(A \text{ or } B)$  equals:

they cannot both occur at the same time

## Choices:

☐  $P(A) + P(B) - P(A \text{ and } B)$

☐  $P(A) \times P(B)$

☒  $P(A) + P(B)$  *correct*

☐  $P(A|B) + P(B|A)$

## Statistics and Probability (STA301)

Question: 7 (Marks: 1)

Attempted Questions: 6

Total Questions: 23

The simultaneous occurrence of two events is called:

## Choices:

☐ Prior probability☐ Subjective probability☐ Conditional probability☒ Joint probability*correct*

## Statistics and Probability (STA301)

Question: 8 (Marks: 1)

Attempted Questions: 7

Total Questions: 23

First moment about origin is always equals to:

.....

## Choices:

☐ Mean☐ Variance☐ Standard Deviation☒ Zero*correct*

## Statistics and Probability (STA301)

Question: 9 (Marks: 1)

Attempted Questions: 8

Total Questions: 23

Which one of the following measurement does *not* divide a set of observations into equal parts?

## Choices:

☐ quartiles☐ deciles☐ percentiles☒ standard deviations*correct*



## Statistics and Probability (STA301)

Question: 10 (Marks: 1)

Attempted Questions: 9

Total Questions: 23

If the first and third quartiles are 22.16 and 56.36 respectively, then the quartile deviation is:

$$Q3-Q1/2$$

## Choices:

17.1

*correct*

34.2

51.3

50.5

## Statistics and Probability (STA301)

Question: 11 (Marks: 1)

Attempted Questions: 10

Total Questions: 23

The height of a student is 60 inches. This is an example of .....?

## Choices:

☐ Continuous data☐ Qualitative data☐ Categorical data☒ Discrete data*correct*

## Statistics and Probability (STA301)

Question: 12 (Marks: 1)

Attempted Questions: 11

Total Questions: 23

Which one of the following statements is true regarding a sample?

## Choices:

☐ It is a part of population

*correct*

☐ It must contain at least five observations

☐ It refers to descriptive statistics

☐ It produces True value

## Statistics and Probability (STA301)

Question: 13 (Marks: 1)

Attempted Questions: 12 Total Questions: 23

Which one of the following graphs is used for a time series data?

## Choices:

☐ Histogram☐ Historigram☐ Frequency curve☐ Frequency polygon

## Statistics and Probability (STA301)

Question: 14 (Marks: 1)

Attempted Questions: 13 Total Questions: 23

Which of the following comes first to make frequency distribution.

## Choices:

☐ No. of Groups☐ Class interval☒ Range*correct*☐ Tally marks

## Statistics and Probability (STA301)

Question: 15 (Marks: 1)

Attempted Questions: 14 Total Questions: 23

The average which is defined as the reciprocal of the arithmetic mean of the reciprocals of the values is called:

## Choices:

☐ Geometric Mean☒ Harmonic Mean*correct*☐ Mode☐ Median

## Statistics and Probability (STA301)

Question: 16 (Marks: 1)

Attempted Questions: 15 Total Questions: 23

If a car is running at the rate of 15km/hr during first 30km, at 20km/hr during the second 30km, which type of average will be used to find the average speed/hr: **page 73**

## Choices:

☐ Weighted Mean☒ Harmonic Mean*correct*☐ Arithematic Mean☐ Geometric Mean



Statistics and Probability (STA301)

Question: **19** (Marks: 2)

Attempted Questions: **23**

Total Questions: **23**

Differentiate between simple and compound events. **page 140**

**SIMPLE & COMPOUND EVENTS**

An event that contains exactly one sample point is defined as a simple event. A compound event contains more than one sample point, and is produced by the union of simple events.

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## Statistics and Probability (STA301)

Question: **22** (Marks: 5)Attempted Questions: **23** Total Questions: **23**

Two events are casted:  $E_1$  is the event that a 6 appears on at least one die,  $E_2$  is the event that a 5 appear on exactly one die. Are  $E_1$  and  $E_2$  independent?

the outcome of one event affects the probability of the other event,  $E_1$  and  $E_2$  are dependent events

## Statistics and Probability (STA301)

Question: **23** (Marks: 5)Attempted Questions: **23**Total Questions: **23**

How many distinct four-digits number can be performed from the following integers 1,2,3,4,5,6 if each integer is used only once?

Total number of distinct four-digit numbers =  $6 * 5 * 4 * 3 = 360$

there are 360 distinct four-digit numbers that can be formed using the integers 1, 2, 3, 4, 5, and 6, where each integer is used only once.

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